

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division
First Semester 2025-2026
Mid-Semester Test
(Answer Key EC-2 Make Up)

Course No. : AIMLCZC418
Course Title : Introduction to Statistical Methods
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 03.01.2026_FN

No. of Pages	= 4
No. of Questions	= 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Answer all the questions:

1. You are a Data Intern at Vertex Auto Parts, reviewing 10-day production output (units/day) post-automation:

190, 250, 240, 270, 230, 280, 245, 265, 285, 380

Provide the five-number summary and identify outliers [3 marks]

Based on the distribution (especially the spread and extreme values), what does this suggest about the stability and consistency of the production process? Is the variation likely due to routine fluctuations, or does it indicate a process shift or data quality concern? Justify briefly. [2 marks]

Solution:

The 10-day production output data (units/day) after automation is:

190, 250, 240, 270, 230, 280, 245, 265, 285, 380

Sorted data: 190, 230, 240, 245, 250, 265, 270, 280, 285, 380

(i) Five-number summary

Q1 position: $0.25 \times (10+1) = 2.75$, between positions 2 (230) and 3 (240):

$Q1 = 230 + 0.75 \times (240 - 230) = 237.5$ (may be different when another method is used)

Median: $(250 + 265)/2 = 257.5$

Q3 position: $0.75 \times (10+1) = 8.25$, between positions 8 (280) and 9 (285):

$Q3 = 280 + 0.25 \times (285 - 280) = 281.25$ (may be different when another method is used)

Minimum = 190, Maximum = 380

Five-number summary: Min=190, Q1=237.5, Median=257.5, Q3=281.25, Max=380 ---- 3M

Outlier detection: $IQR = 281.25 - 237.5 = 43.75$ (may be different when another method is used)

Upper bound = $281.25 + 1.5 \times 43.75 = 346.875$

Since $380 > 346.875$, 380 is an outlier.

(ii) Process stability and consistency assessment

Most production days range between 190 and 285 units, which suggests generally stable daily output. The 380-unit day is far above the normal range and is a statistical outlier.

This extreme value is not due to routine daily fluctuations.

Possible reasons include a temporary process improvement, such as an extra shift or higher efficiency, or a data recording error like double counting. Overall, the outlier indicates a special cause event. The process is mostly stable but requires investigation into the high-output day before confirming it is fully in control. -----2M

2. The probabilities that an academician, a businessman, or a politician is appointed Vice-Chancellor of a university are 0.5, 0.3 and 0.2 respectively. The probabilities that research work is promoted given the appointment of each are:
 Academician: 0.8, Businessman: 0.6, Politician: 0.4. It is observed that research work has been promoted. Find the probability that the Vice-Chancellor is an academician.
 [5 marks]

Solution:

Let:

A = event that an academician is appointed VC

B = event that a businessman is appointed VC

P = event that a politician is appointed VC

Given:

$$P(A) = 0.5, P(B) = 0.3, P(P) = 0.2$$

Let R = event that research work is promoted.

Given:

$$P(R|A) = 0.8, P(R|B) = 0.6, P(R|P) = 0.4 \text{ -----2M}$$

We want $P(A|R)$.

Using Bayes' theorem:

$$P(A|R) = P(R|A) * P(A) / P(R)$$

Compute $P(R)$ by total probability:

$$\begin{aligned} P(R) &= P(R|A)P(A) + P(R|B)P(B) + P(R|P)P(P) \\ &= (0.8)(0.5) + (0.6)(0.3) + (0.4)(0.2) \\ &= 0.4 + 0.18 + 0.08 \\ &= 0.66 \text{ -----2M} \end{aligned}$$

Now:

$$\begin{aligned} P(A|R) &= \frac{P(R|A)P(A)}{P(R)} \\ &= \frac{0.8 * 0.5}{0.66} \\ &= \frac{0.4}{0.66} \\ &= 40/66 \\ &= 20/33 = 0.6060 \text{ (Approx.) ----- 1M} \end{aligned}$$

3. The final exam scores for ISM are modeled as normally distributed with mean 73 and standard deviation 8. A teaching assistant makes the following claims:

- a) Most students scored below 91.
- b) Scoring between 65 and 89 puts you in the central majority.
- c) Only 5% of students scored above a certain threshold — that's the cut-off for the Dean's List. Using the model:

(i) Verify the first claim: Find $P(X < 91)$. Is it indeed “most”?

[2 marks]

(ii) Quantify the second claim: Find $P(65 < X < 89)$

[2 marks]

(iii) Determine the exact cut-off score for the Dean's List.

[1 mark]

Solution:

Exam scores $X \sim N(\mu = 73, \sigma = 8)$

(i) Verify “Most students scored below 91”

$$z = \frac{91 - 73}{8} = 2.25$$

From z-table, $P(Z < 2.25) = 0.9878$. Thus, $P(X < 91) = 0.9878$. Claim is true.

(ii) Verify “Scoring 65 to 89 is central majority”

$$z_{65} = \frac{65 - 73}{8} = -1, \quad z_{89} = \frac{89 - 73}{8} = 2$$

$$P(65 < X < 89) = P(-1 < Z < 2) = \Phi(2) - \Phi(-1)$$

From tables, $\Phi(2) = 0.9772$, $\Phi(-1) = 0.1587$:

$$P(65 < X < 89) = 0.9772 - 0.1587 = 0.8185$$

81.85% in this range → central majority. Claim is true.

(iii) Dean's List cut-off (top 5%)

For top 5%, $P(X > x) = 0.05$, so $P(X < x) = 0.95$.

From z-table, $z_{0.95} = 1.645$:

$$x = 73 + 1.645 \times 8 = 73 + 13.16 = 86.16$$

Dean's List cut-off = **86.2** marks.

Hence,

$$P(X < 91) = 0.9878$$

$$P(65 < X < 89) = 0.8185$$

$$\text{Cut-off} = \mathbf{86.2}$$

4. A cybersecurity operations team models joint behaviour of two discrete metrics during night shifts:

Let X = number of servers compromised (0, 1, 2, 3),

Y = severity level of the attack (values: -1, 0, 1, 2). where (X, Y) has joint probability mass function $f(x, y) = c(x^2 + y + 1)$ for $x \in \{0, 1, 2, 3\}$, $y \in \{-1, 0, 1, 2\}$

(Note: The domain includes negative severity — interpreted as “false alarm” or “low-impact probe.”)

- Find the normalizing constant c [1 mark]
- Compute $P(X = 1, Y \geq 0)$ [1 mark]
- Are X and Y independent? Justify using marginal distributions. [1 mark]
- Given that at least one server was compromised ($X \geq 1$), what is the probability that the attack severity was non-negative ($Y \geq 0$)? [2 marks]

Soln:- Given $f(x, y) = c(x^2 + y + 1)$ for $x \in \{0, 1, 2, 3\}$
 $y \in \{-1, 0, 1, 2\}$.

Here X — no. of servers compromised.

Y — severity level of the attack.

a) For finding normalised constant c ,

$$\sum_{x=0}^3 \sum_{y=-1}^2 f(x, y) = 1 \Rightarrow \sum_{x=0}^3 \sum_{y=-1}^2 c(x^2 + y + 1) = 1$$

$$\begin{aligned} \Rightarrow & [f(0, -1) + f(0, 0) + f(0, 1) + f(0, 2)] \\ & + [f(1, -1) + f(1, 0) + f(1, 1) + f(1, 2)] \\ & + [f(2, -1) + f(2, 0) + f(2, 1) + f(2, 2)] \\ & + [f(3, -1) + f(3, 0) + f(3, 1) + f(3, 2)] = 1 \end{aligned}$$

$$c \left[[0+1+2+3] + [1+2+3+4] + [4+5+6+7] + [9+10+11+12] \right] = 1$$

$$\Rightarrow 80c = 1 \Rightarrow \boxed{c = \frac{1}{80}} \quad (\text{Mark})$$

$$\begin{aligned} \text{b). } P(x=1, y \geq 0) &= \sum_{y=0}^2 c(i^2 + y + 1) \quad \because x=1 \\ &= \frac{1}{80} [(1+0+1) + (1+1+1) + (1+2+1)] \\ &= \frac{9}{80} \end{aligned}$$

$$\therefore \boxed{P(x=1, y \geq 0) = \frac{9}{80}} \quad (\text{Mark}).$$

$$\text{c). } P(x=\tilde{x}) = c \sum_{y=-1}^2 (\tilde{x}^2 + y + 1)$$

$$= c [\tilde{x}^2 + (\tilde{x}^2 + 1) + (\tilde{x}^2 + 2) + (\tilde{x}^2 + 3)]$$

$$P(Y=y) = c \sum_{\tilde{x}=0}^3 (\tilde{x}^2 + y + 1)$$

$$= c [(0^2 + y + 1) + (1^2 + y + 1) + (2^2 + y + 1) + (3^2 + y + 1)]$$

Let us take an exmple of $x=1, y=0$, we get

$$P(x=1, y=0) = \frac{1}{80} [1+1] = \frac{1}{40} \quad \therefore c = \frac{1}{80}$$

$$P(X=1, Y=0) = \frac{1}{80} [1+1] = \frac{1}{40} \quad \because C = \frac{1}{80}$$

$$\text{But } P(X=1) = \frac{1}{80} (1+2+3+4) = \frac{10}{80}$$

$$\& P(Y=0) = \frac{1}{80} (1+2+5+10) = \frac{18}{80}$$

$$P(X=1) \cdot P(Y=0) = \frac{10}{80} \times \frac{18}{80} = \frac{9}{320} \neq$$

$$P(X=1, Y=0)$$

Hence X & Y are not independent.
(1 Mark)

$$d) P(Y \geq 0 / X \geq 1) = \frac{P(X \geq 1, Y \geq 0)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - P(X=0) = 1 - \frac{1}{80} (0+1+2+3)$$

$$= 1 - \frac{6}{80}$$

$$= \frac{74}{80}$$

$$= \frac{37}{40}$$

$$\& P(X \geq 1, Y \geq 0) = f(1,0) + f(2,0) + f(3,0) \\ + f(1,1) + f(2,1) + f(3,1) \\ + f(1,2) + f(2,2) + f(3,2)$$

$$= c \left[(2+5+10) + (3+6+11) + (4+7+12) \right]$$

$$= \frac{1}{80} [60]$$

$$= \frac{3}{4}$$

$$\therefore P(Y \geq 0 / X \geq 1) = \frac{3/4}{37/40} = \frac{30}{37} \approx 81\%.$$

(2 Marks)

5. A data centre monitors the normalized overheating index X (unitless, range: 0 to 3). The cumulative distribution function (CDF) of X is:

$$F(x) = \begin{cases} 0, & x \leq 0 \\ kx^2, & 0 < x \leq 1 \\ k(2x - 1), & 1 < x \leq 3 \\ 1, & x > 3 \end{cases}$$

$X=0$ means fully cooled; $X=3$ triggers emergency shutdown.

- Find the constant k [2 marks]
- Find the probability that the system is in warning zone (i.e., $1 \leq x \leq 2$). [2 marks]
- Compute the probability density function $f(x)$ [1 mark]

5) Soln. WKT. $\frac{d}{dx} F(x) = f(x)$.

$$\therefore f(x) = \begin{cases} 0 & x \leq 0 \\ 2Kx & 0 < x \leq 1 \\ 2K & 1 < x \leq 3 \\ 0 & x > 3 \end{cases}$$

a) $f(x)$ is pdf, then $f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{-\infty}^0 0 dx + \int_0^1 2Kx dx + \int_1^3 2K dx + \int_3^{\infty} 0 dx = 1$$

$$\Rightarrow K(x^2)_0^1 + 2K(x)_1^3 + 0 = 1$$

$$\Rightarrow K + 2K(3-1) = 1$$

$$\Rightarrow K + 4K = 1$$

$$\Rightarrow K + 4K = 1$$

$$\Rightarrow \boxed{K = \frac{1}{5}}$$

(2 Mark)

b) $P(1 \leq x \leq 2) = F(2) - F(1)$.

$$= K(2(2)-1) - K(1)^2$$

$$= K[3-1]$$

$$= \frac{1}{5}(2)$$

(2 marks)

c) $f(x) = \begin{cases} 2Kx & 0 \leq x \leq 1 \\ 2K & 1 < x \leq 3 \\ 0 & \text{otherwise} \end{cases}$

$$= \begin{cases} \frac{2}{5}x & 0 < x \leq 1 \\ \frac{2}{5} & 1 < x \leq 3 \\ 0 & \text{otherwise} \end{cases} \quad (1 \text{ mark})$$

6. a) At Secure Bank, historical data shows that cheque-processing errors (e.g., amount mismatch, signature miss) occur randomly and independently at an average rate of 1.2 errors per 100 cheques processed. On a typical day, the teller processes 250 cheques. Let X = number of errors observed in a day.

(i) Specify the distribution of X , including its parameter. [1 mark]

(ii) Find the probability that at most 2 errors occur in a day. [1 mark]

(iii) What is the probability of observing more than 4 errors? [1 mark]

6a) Soln. Given mean rate for 100 cheques = 1.2 errors.

(i) For 250 cheques, $\lambda = 1.2 \times \frac{250}{100} = 3$.

So, $X \sim \text{Poisson}(\lambda=3)$. (1 mark).

ii). $P(\text{At most 2 errors per in a day})$

$$= P(X \leq 2) = P(X=0) + P(X=1) + P(X=2).$$

$$= \frac{e^{-3} 3^0}{0!} + \frac{e^{-3} 3^1}{1!} + \frac{e^{-3} 3^2}{2!}$$

$$= e^{-3} \left[1 + 3 + \frac{9}{2} \right]$$

$$= 0.0498 \times 8.5$$

$$= 0.4233. \quad (1 \text{ mark}).$$

iii). $P(\text{more than 4 errors}) = P(X > 4) = 1 - P(X \leq 4)$

$$= 1 - \left[e^{-3} (1 + 3 + 4.5 + 4.5 + 3.375) \right]$$

$$= 1 - 0.8153$$

$$= 0.1847.$$

(1 mark).

b) At Bharat Auto Forge, every brake disc undergoes inspection for three common manufacturing flaws:

W: Warpage (non-flat surface),

C: Cracking (micro-fractures),

S: Surface scoring (tool-mark scratches).

A recent audit of 10,000 units revealed: 1,200 had warpage, 850 had cracking, 1,500 had surface scoring, 300 had both warpage and cracking, 400 had both warpage and scoring, 250 had both cracking and scoring, 100 had all three defects

(i) What is the probability that a randomly selected brake disc has at least one of these three defects? [1 mark]

(ii) How many discs (out of 10,000) are expected to be completely defect-free? [1 mark]

6b) Soln:

$$n(W) = 1200, n(C) = 850, n(S) = 1500$$

$$n(W \cap C) = 300, n(W \cap S) = 400, n(C \cap S) = 250$$

$$n(W \cap C \cap S) = 100 \text{ out of } 10,000$$

$$(i) \cdot P(\text{At least one of three defects}) = P(W \cup C \cup S)$$

$$= P(W) + P(C) + P(S) - P(W \cap C) - P(C \cap S) - P(S \cap W) + P(W \cap C \cap S)$$

$$= \frac{2700}{10000}$$

$$= 0.27$$

(1 Mark)

(ii) No. of discs out of 10,000 that are expected to be completely defect free

$$P (1 - 0.27) \times 10000$$

$$= 7300$$

(1 Mark)